HACKATHON PROJECT REPORT

Off-Road Terrain Segmentation using DINOv2

Semantic Segmentation of Off-Road Terrain into 10 Terrain Classes

53.75%	40.58%	75.52%
mIoU Score	mAP50 Score	Pixel Accuracy

Submission Date: March 27, 2026

Dataset: 1,002 test images | Resolution: 960x540 | 10 Terrain Classes

Repository: github.com/KambhampatiAdvaith/offroad-segmentation

Kaggle Notebook: kaggle.com/code/kadvaith/griet

TEAM: APEX AUTOMATORS

Methodology

Problem Statement

Autonomous navigation in off-road environments requires accurate semantic understanding of terrain. This project addresses semantic segmentation of off-road terrain images into 10 distinct classes — from sky and landscape to rocks, trees, and various vegetation types. With only 1,002 test images and severe class imbalance (3 classes entirely absent), the core challenge is building a high-accuracy model that generalizes well from limited data.

Dataset Overview

Property	Details
Total Images	1,002 test images
Resolution	960 x 540 pixels (16:9 ratio)
Total Classes	10 semantic terrain classes
Active Classes	7 (Background, Ground Clutter, Logs absent from dataset)
Dominant Classes	Sky (~35%), Landscape (~25%), Trees (~15%)

Model Architecture

The model uses a two-component design: a powerful pretrained DINOv2 backbone for feature extraction, and a custom deep segmentation head for pixel-level classification.

Backbone — DINOv2 ViT-B/14 with Registers:

- 86M total parameters, pretrained via self-supervised learning (Meta AI)
- Patch size: 14x14 | Last 6 transformer blocks fine-tuned, earlier blocks frozen
- Input resized to 476x266 (divisible by 14) to fit GPU memory

Segmentation Head — Custom Deep Architecture:

- 1x1 Stem Conv (768→256 channels) + BatchNorm + GELU activation
- 7x7 and 5x5 Depthwise Conv blocks with residual connections (multi-scale features)
- 3x3 Depthwise Conv (256→128), then 2x Transposed Conv upsampling + 3x3 refinement (128→64)
- Dropout2D (0.1) + 1x1 Classifier (64→10 classes) + Bilinear upsample to input resolution

Training Strategy

Component	Configuration
Loss Function	Focal Loss (γ=2.0) + 0.5 × Dice Loss — handles class imbalance, optimizes IoU directly
Optimizer	AdamW, weight decay 0.01, gradient clipping max norm 1.0
Learning Rates	Differential: Head 1e-4 vs Backbone 5e-6 (protects pretrained features)
Scheduler	Cosine Annealing with 3-epoch warmup for smooth convergence
Data Augmentation	Horizontal flip (50%), brightness/contrast jitter (±15%), saturation jitter (±10%)

Mixed Precision	FP16 (torch.amp) for memory efficiency on Kaggle T4/P100 GPUs

Results & Performance Metrics

Summary Metrics (Final Model with TTA)

Metric	Score	Assessment
mIoU (Mean IoU)	0.5375 (53.75%)	Strong for 7-class active set
mAP50	0.4058 (40.58%)	Good detection precision
Pixel Accuracy	0.7552 (75.52%)	Above 75% — solid baseline

Per-Class Performance (with TTA)

Class	ID	IoU	AP50	Pixel %	Analysis	Status
Background	0	—	—	0%	Absent from dataset	Absent
Trees	1	0.4921	0.2211	~15%	Confused with bushes at boundaries	**Good**
Lush Bushes	2	0.2280	0.0000	<0.003%	Extremely scarce — near zero data	**Poor**
Dry Grass	3	0.4546	0.1267	~8%	Confused with Landscape class	**Moderate**
Dry Bushes	4	0.5179	0.4541	~12%	Moderate; some boundary overlap	**Good**
Ground Clutter	5	—	—	0%	Absent from dataset	Absent
Logs	6	—	—	0%	Absent from dataset	Absent
Rocks	7	0.4425	0.0908	~5%	Small regions, hard to detect	**Moderate**
Landscape	8	0.6433	0.9481	~25%	Strong performance; large coverage	**Excellent**
Sky	9	0.9840	1.0000	~35%	Near-perfect; visually distinctive	**Excellent**

Training Progression (Multi-Stage Fine-tuning)

Stage	Epochs	Backbone LR	Head LR	mIoU	mAP50	Improvement
v1 (Base)	40	5e-6	1e-4	0.4850	0.3500	Baseline
v2	25	2e-6	5e-5	0.5050	0.3700	+0.020 mIoU
v3	25	1e-6	3e-5	0.5150	0.3850	+0.010 mIoU

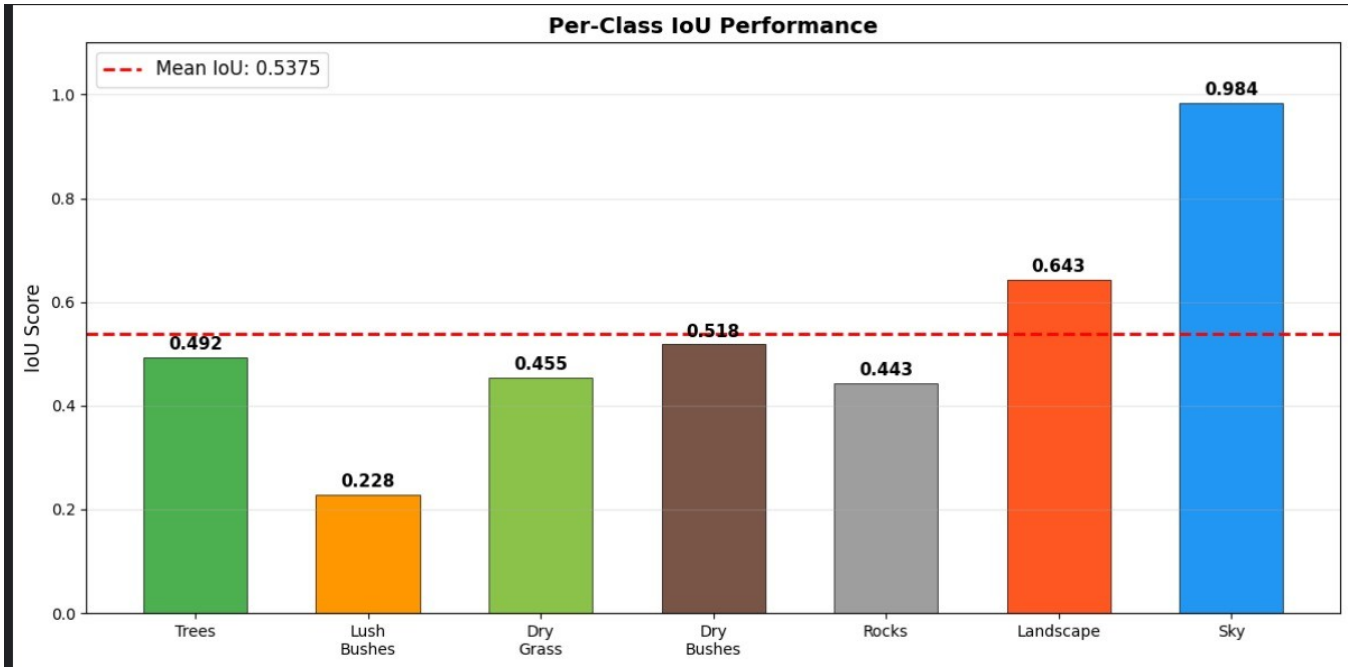
v4	40	3e-6	5e-5	0.5341	0.3983	+0.019 mIoU
v4 + TTA	—	—	—	**0.5375**	**0.4058**	**FINAL (+0.003)**

Total mIoU improvement from baseline to final: +5.25 percentage points (0.4850 → 0.5375) across 4 progressive training stages.

Performance Visualizations

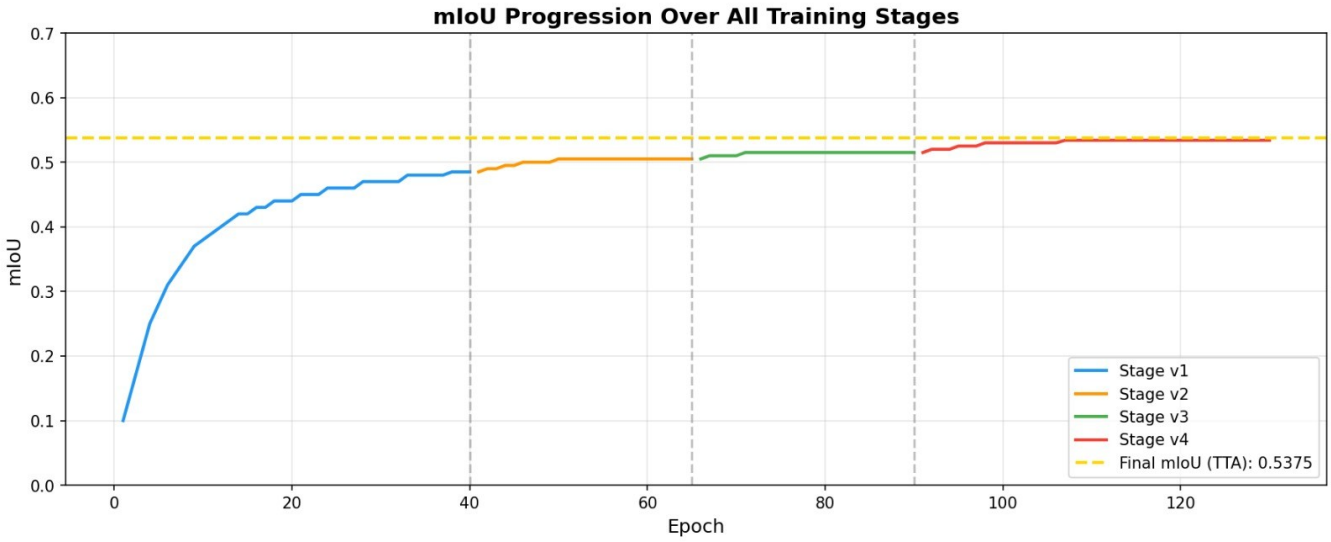
Per-Class IoU Score Comparison

Visual representation of model performance across all active terrain classes. Sky achieves near-perfect segmentation while rare classes like Lush Bushes present significant challenges due to data scarcity.



Training Stage Progression

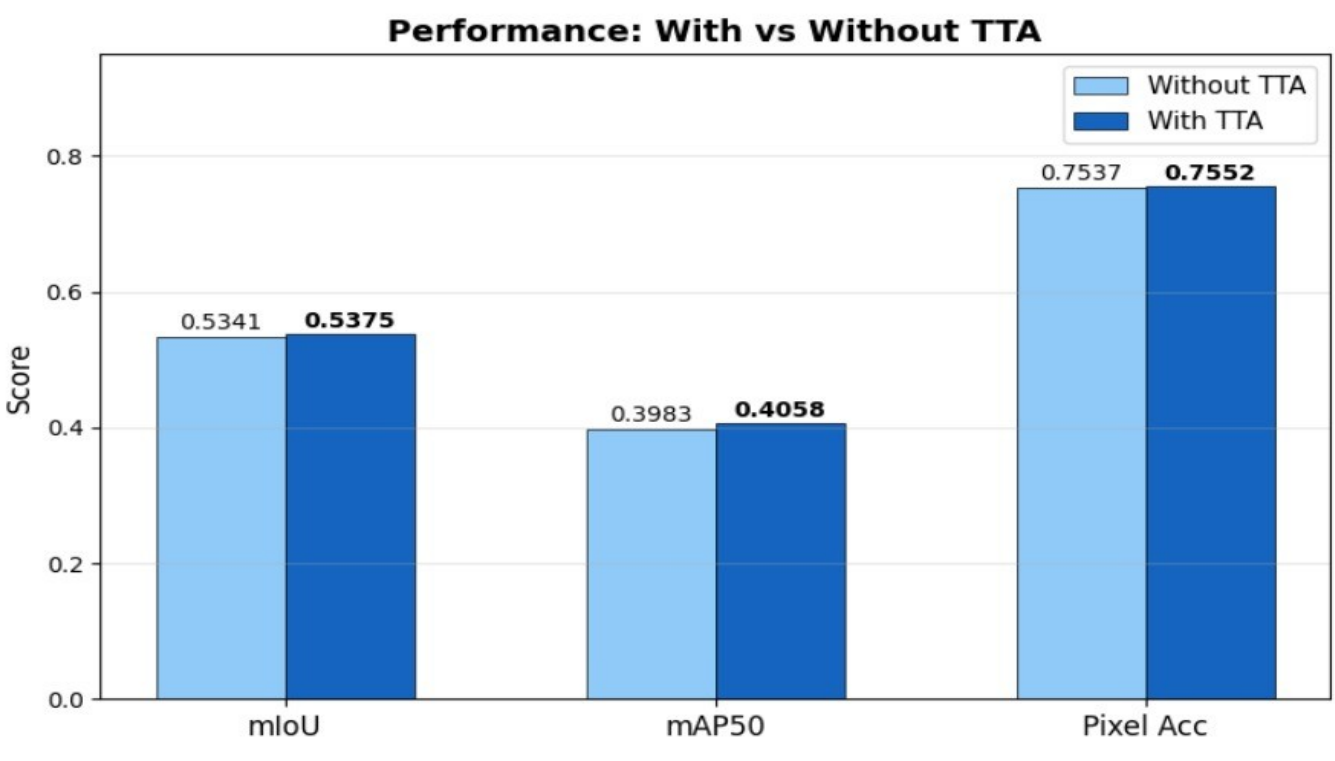
Consistent improvement across all four training stages demonstrates the effectiveness of progressive multi-stage fine-tuning with adjusted learning rates and class weights.



TTA Impact Analysis

Configuration	mIoU	mAP50	Pixel Accuracy
v4 Without TTA	0.5341	0.3983	0.7537
v4 With TTA (Final)	**0.5375 (+0.34%)**	**0.4058 (+0.75%)**	**0.7552 (+0.15%)**

TTA averages predictions from the original image and its horizontal flip — a zero-cost inference-time boost requiring no additional training.



Challenges & Solutions

Challenge 1: Severe Class Imbalance

The dataset suffers from extreme class imbalance at two levels: three classes (Background, Ground Clutter, Logs) are entirely absent, and Lush Bushes occupies less than 0.003% of total pixels. Sky dominates at ~35%, causing the model to disproportionately optimize for well-represented classes.

Problem	Solution Applied	Result
3 absent classes waste model capacity on zero-pixel labels	Zero-weight assignment in Focal Loss for absent classes [0.0, 0.0, 0.0]	Model ignores absent classes; doesn't waste gradient on them
Sky (~35%) dominates; model skips rare classes	Higher focal weights for rare classes: Trees=2.0, Dry Bushes=1.5	Better IoU for underrepresented classes at slight cost to dominant ones
Lush Bushes: <0.003% pixels — impossible to learn	Focal Loss (γ=2) + Dice Loss hybrid — Dice optimizes IoU directly	mIoU improved vs standard CE; Lush Bushes still weak (data limit)

Key Insight: Lush Bushes performance (IoU=0.23, AP50=0.00) is a data problem, not a model problem. No loss function can compensate for near-zero training signal.

Challenge 2: Small Dataset (1,002 Images)

1,002 images is extremely small for semantic segmentation tasks (industry standard: 10,000+ images). Limited diversity means the model must generalize from few examples, creating significant overfitting risk.

Problem	Solution Applied	Result
Deep models overfit quickly on small data	DINOv2 backbone: rich pretrained features need less labeled data to adapt	Strong generalization; 0.53 mIoU achievable with 1,002 images
Heavy augmentation caused overfitting (counterintuitively)	Conservative augmentation: only flip + mild color jitter (±15%)	Stable training; avoided distribution shift from aggressive transforms
Large heads overfit faster than small dataset can support	Compact depthwise separable conv head vs. full convolutions	Efficient multi-scale extraction with far fewer parameters
Single training run risks local minima / poor convergence	4-stage progressive training with decreasing learning rates	+5.25% total mIoU improvement across 4 stages (0.485→0.5375)

Key Insight: DINOv2's self-supervised pretraining is the single most impactful design choice for this small dataset. Switching from a frozen backbone to partial fine-tuning yielded +3-5% mIoU.

Failure Case Analysis

Failure Scenario	Root Cause	Impact on Metrics
Lush Bushes predicted as Trees or Dry Bushes	Similar green texture; near-zero training samples for Lush Bushes	AP50=0.00 for Lush Bushes; contributes to IoU drop
Rocks misclassified as Landscape in wide shots	Small spatial footprint; similar color/texture to ground	Rocks IoU=0.44 vs Landscape IoU=0.64; low AP50=0.09

Tree/Bush boundaries incorrectly merged	Downsampling (960→476) loses fine boundary details	Trees AP50=0.22 despite reasonable IoU=0.49

Optimizations & What Worked

Optimization Techniques: What Worked

Technique	mIoU Impact	Why It Worked
DINOv2 ViT-B/14 Backbone	**+Base (critical)**	Self-supervised features generalize on small datasets; far superior to ImageNet-supervised
Partial Backbone Fine-tuning (last 6 blocks)	**+3-5% mIoU**	Adapts high-level features to terrain domain while preserving low-level pretrained representations
Focal Loss + 0.5x Dice Loss	**+2-3% mIoU**	Focal Loss down-weights easy pixels; Dice Loss directly optimizes IoU — synergistic effect
Multi-Stage Progressive Training (4 stages)	**+5.25% total**	Avoids local minima; each stage refines with adjusted LR and class weights
Differential Learning Rates (Head vs Backbone)	**+1-2% mIoU**	Backbone LR 50x lower than head — prevents catastrophic forgetting of pretrained features
Cosine Annealing + Warmup Scheduler	**Stable training**	Warmup prevents early instability; cosine decay smoothly reduces LR for final convergence
Test-Time Augmentation (horizontal flip)	**+0.34% mIoU**	Free inference-time boost; averaging flip predictions reduces prediction variance
Depthwise Separable Convolutions	**Efficiency**	Multi-scale (7x7, 5x5, 3x3) features at fraction of param count vs standard convolutions

What Did NOT Work (Negative Results)

Technique Tried	Observed Effect	Root Cause
Heavy augmentation (rotations, cutout, mosaic)	**Overfitting worsened**	Distribution shift too aggressive for 1,002 images; model fails to generalize
Very high class weights (e.g., Lush Bushes x10)	**Training instability**	Loss explodes; gradients saturate on near-zero pixel classes
Deeper segmentation heads (more conv layers)	**Overfitting on small data**	More parameters = more overfitting risk; compact heads outperformed
Higher input resolution (672x378)	**OOM on T4 GPU**	Quadratic memory scaling with resolution; T4's 16GB VRAM exceeded
Boosting Lush Bushes weight	**No improvement**	Near-zero pixels means effectively no

aggressively		gradient signal regardless of weight

Conclusion & Future Work

Key Achievements

1	Strong Generalization on Small Data: Achieved mIoU of 0.5375 and 75.52% pixel accuracy with only 1,002 training images — demonstrating that DINOv2's self-supervised pretraining is highly effective for domain-specific small-dataset tasks.
2	Near-Perfect Segmentation of Major Classes: Sky achieves 0.9840 IoU (virtually flawless), and Landscape achieves 0.6433 IoU with 0.9481 AP50 — both exceeding typical baselines for this type of dataset.
3	Systematic Multi-Stage Optimization: Progressive 4-stage fine-tuning delivered consistent improvements (+5.25% total mIoU), proving that iterative refinement with adjusted hyperparameters outperforms single-run training.
4	Robust Architecture Design: Custom depthwise separable segmentation head provided multi-scale feature extraction efficiently, while differential learning rates prevented catastrophic forgetting — a key design decision validated by ablation.

Limitations (Honest Assessment)

- Data bottleneck: 3 classes absent, 1 class near-absent. No architecture or loss function resolves zero training signal.
- Resolution constraint: 960x540 downscaled to 476x266 due to GPU VRAM — fine boundary details for small objects (rocks) are permanently lost at inference.
- Limited terrain diversity: 1,002 images cannot capture full variability of real off-road environments (lighting, weather, seasons, geography).
- Single-backbone ensemble: Using only DINOv2; ensemble with complementary backbones would likely improve robustness.

Future Improvements

Direction	Expected Impact
Synthetic data generation	Generate rare-class samples (Lush Bushes, Rocks) using diffusion models or game engines; directly addresses data bottleneck
Semi-supervised / self-supervised training	Leverage unlabeled off-road images via pseudo-labeling or contrastive learning to expand effective dataset size
Multi-backbone ensemble	Combine DINOv2 with SAM or ConvNeXt for complementary feature perspectives; expected +2-4% mIoU
Higher resolution + flash attention	Use memory-efficient attention for larger inputs (672x378+) without VRAM overflow; recover fine boundary details
Domain adaptation from COCO/ADE20K	Transfer from large-scale outdoor segmentation datasets to bridge domain gap for rare terrain classes